Topology Optimization of Chip-Scale Quantum-Comb Inertial Sensors: A Differentiable Fisher-Information Framework from the Keldysh Path Integral

Companion manuscript to "Keldysh Path Integral Quantization of Nonlinear Open Nanophotonic Structures via Quasi-Normal Modes"

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Abstract

We extend the Keldysh quasi-normal-mode (QNM) quantization framework of the companion manuscript to the design and topology optimization of ultra-compact, on-chip quantum-comb inertial sensors. The target device is a freeform nonlinear active nanophotonic structure operating in the comb-lasing regime with rotational/inertial sensing of an output free-space comb via spatio-temporal photon-statistics measurements.

We derive: (i) the Keldysh action for a multi-mode $\chi^{(3)}$ nonlinear active cavity with rotational (Sagnac) coupling; (ii) the above-threshold mean-field saddle (a generalised Lugiato–Lefever soliton-comb solution with classical CW/CCW splitting); (iii) the Bogoliubov problem for linear quantum fluctuations around the saddle, showing each comb-pair $\{+m, -m\}$ reduces to a two-mode squeezed amplifier with Sagnac splitting; (iv) the input–output relation projecting cavity QNM fluctuations onto the free-space output field at detector pixels $\mathbf{r}_{\mathrm{det}}$; (v) the closed-form classical Fisher information (FI) of a heterodyne or photon-counting measurement on the comb-pair output for the Sagnac parameter Ω; and (vi) the adjoint gradient of this FI with respect to the four design fields $\{\varepsilon, g_{\mathrm{gain}}, \chi^{(2)}, \chi^{(3)}\}(\mathbf{r})$, delivering a fully differentiable TopOpt objective.

We benchmark the linearised comb-pair covariance against direct qutip Lindblad simulation, finding $< 1\%$ agreement in the twin-beam squeezed quadrature variance up to $S = 2\chi/\kappa = 0.85$. A worked example for a GaAs micro-ring at $\lambda = 1550\,\mathrm{nm}$, $Q = 10^5$, $R = 10\,\mu\mathrm{m}$ predicts a Sagnac shot-noise limit of $\sim 10^{-7}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ at the single-comb-pair level, with quantum-enhancement up to $-3\,\mathrm{dB}$ intracavity squeezing per pair.

We close with three open derivations not yet in this draft (4-point spatio-temporal correlator, generating-functional Fisher information for photon-counting POVMs, and adjoint method for higher-moment objectives), which the framework supports architecturally but which we do not work out exhaustively here.

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1 Setting and target observable

1.1 Device architecture

Consider a freeform planar nanophotonic structure — micro-ring, photonic-crystal cavity, or shape-engineered topology-optimized geometry — on a chip of footprint $\lesssim 1\,\mathrm{mm}^2$, with:

- **Material:** a $\chi^{(3)}$ Kerr-active substrate (Si_3N_4, AlGaAs, AlN, $LiNbO_3$, or topology-optimized composite) hosting a $\chi^{(2)}$ active layer (epitaxial GaAs/AlGaAs for d_{14}, or $LiNbO_3$ for d_{33}).
- **Gain region:** embedded quantum-well or quantum-dot active medium providing optical gain with saturation, modelled as in the gain section of the companion manuscript.
- **Cavity Q:** 10^4–10^6 at the comb operating wavelength.
- **Output:** free-space radiation through one or more out-coupling channels; detected by a multi-pixel photon-counting or heterodyne array at positions $\{\mathbf{r}_{\mathrm{det}}^{(p)}\}$.

The device operates in the *comb-lasing regime*: a single narrow-linewidth pump tone drives the gain-loaded cavity above threshold; four-wave mixing (FWM) generates a phase-locked frequency comb with $2N+1$ lines $\{\omega_m : m = -N, \ldots, N\}$ at FSR spacing $\Delta\omega_{\mathrm{FSR}}$, output as a coherent free-space comb.

1.2 Inertial parameter and output observable

The sensing parameter is the rotational rate Ω (gyroscope) or, generalising, an acceleration $\mathbf{a}$ that couples to the photonic mode through a small mechanical displacement of the cavity boundary. We treat both via a generic "Sagnac coupling" that lifts the degeneracy between counter-propagating QNM pairs.

The output observable is built from the spatio-temporal output field $\hat{\mathbf{E}}^{\mathrm{out}}(\mathbf{r}_{\mathrm{det}}^{(p)}, t)$, projected onto a measurement basis (heterodyne quadratures, photon-number counts, or higher-order coincidences). The figure of merit is the **classical Fisher information** $F(\Omega;\theta)$ of the measurement outcome distribution w.r.t. Ω, evaluated as a function of the design parameters θ (continuum permittivity field $\varepsilon(\mathbf{r})$ or its TopOpt parameterisation). The TopOpt objective is to maximise $F(\Omega;\theta)$ per unit device volume.

2 Keldysh action with rotational coupling

2.1 Linear part

The Keldysh action for the QNM amplitudes $a_\lambda^\pm(t)$ of the cavity (derived in detail in the companion manuscript, Secs. 5–7) is

$$S_{\mathrm{K}}^{\mathrm{lin}} = \int \frac{d\omega}{2\pi} \begin{pmatrix} a^{\mathrm{cl}} & a^{\mathrm{q}} \end{pmatrix}^* \begin{pmatrix} 0 & [G^A]^{-1} \\ [G^R]^{-1} & D^{\mathrm{K}} \end{pmatrix} \begin{pmatrix} a^{\mathrm{cl}} \\ a^{\mathrm{q}} \end{pmatrix}, \tag{1}$$

with $G^R(\omega) = (\omega - \tilde{\omega}_\lambda)^{-1}$, $\tilde{\omega}_\lambda = \omega_\lambda - i\kappa_\lambda/2 + ig_\lambda/2$ including both passive loss κ_λ and active gain g_λ, and Keldysh component $D^{\mathrm{K}} = -i\kappa_\lambda(2n_\lambda + 1) + ig_\lambda(2N_{\mathrm{sp}} + 1)$. The Cholesky-dressed multi-mode operators $\hat{a}_\nu = (L^\dagger \hat{q})_\nu$ satisfy $[\hat{a}_\nu, \hat{a}_{\nu'}^\dagger] = \delta_{\nu\nu'}$ exactly within the kept N-mode subspace.

2.2 Sagnac coupling: rotational frame analysis

In a frame rotating with angular velocity $\mathbf{\Omega}$, the Maxwell equations acquire a non-reciprocal term $+2(\mathbf{\Omega} \times \mathbf{r}) \cdot \nabla$ acting on the electromagnetic field [1, 2]. At the level of the QNM eigenvalue problem, this is a first-order perturbation that splits the eigenvalues of any pair of counter-propagating modes $\{\tilde{\mathbf{f}}_m, \tilde{\mathbf{f}}_{-m}\}$ (e.g. azimuthal $e^{\pm im\phi}$ modes of a ring resonator) by a Sagnac frequency:

$$\tilde{\omega}_{\pm m}^{(\Omega)} = \tilde{\omega}_m \pm \Omega_{\mathrm{eff},m}, \quad \Omega_{\mathrm{eff},m} = \frac{4A_{\mathrm{eff},m}\,\omega_m}{n_g c\, L_{\mathrm{eff}}}\,\Omega \approx \frac{A_{\mathrm{eff},m}\,\omega_m}{c^2}\,\Omega, \tag{2}$$

where $A_{\mathrm{eff},m}$ is the geometric area enclosed by the mode and L_{eff} its optical path length. For an azimuthal mode with intensity $I_m(\mathbf{r})$,

$$A_{\mathrm{eff},m} = \frac{1}{2} \int d^3r\, (\mathbf{r} \times \hat{z}) \cdot \mathrm{Im}\big[\tilde{\mathbf{f}}_m^* \nabla \tilde{\mathbf{f}}_m\big] \Big/ \int d^3r\, \varepsilon |\tilde{\mathbf{f}}_m|^2. \tag{3}$$

For a freeform geometry, $A_{\mathrm{eff},m}$ is computed directly from the QNM mode function obtained by FDTD.

The Sagnac coupling enters the Keldysh action additively:

$$S_{\mathrm{Sag}} = \sum_m \int \frac{d\omega}{2\pi}\, \Omega_{\mathrm{eff},m} \big[a_{+m}^*\, a_{+m} - a_{-m}^*\, a_{-m}\big]_K, \tag{4}$$

where the bracket denotes the standard Keldysh-rotated form. This is a *linear in* Ω perturbation: classical, real-valued, and fully differentiable w.r.t. both Ω and the geometry (through $A_{\mathrm{eff},m}[\varepsilon(\mathbf{r})]$).

2.3 Nonlinear vertices

The $\chi^{(3)}$ four-wave mixing vertex couples four comb modes:

$$S_{\mathrm{NL}}^{(3),\mathrm{K}} = \sum_{mnpq} \hbar g_{mnpq}^{(3)} \int dt \Big[\big(a_m^+\big)^* \big(a_n^+\big)^* a_p^+ a_q^+ - (+ \to -)\Big]\, \delta(\omega_m + \omega_n - \omega_p - \omega_q). \tag{5}$$

Energy conservation enforces $m + n = p + q$ for equally-spaced comb lines. The dominant FWM process for comb generation is the *degenerate* pump-driven pair production $2\omega_0 \to \omega_{+m} + \omega_{-m}$, with vertex amplitude $g_{m,-m,0,0}^{(3)}$. The Kleinman reality theorem of the companion manuscript (Sec. 9) ensures $g_{m,-m,0,0}^{(3)} \in \mathbb{R}$ for passive Kleinman $\chi^{(3)}$.

2.4 Gain saturation and pump driving

The active medium is included via the gain-modified noise kernel and a saturation $\chi^{(3)}_{\rm sat} \in i\mathbb{R}$ (Sec. 16 of the companion manuscript). Above lasing threshold, the saturated gain clamps to the loss: $g^{\rm sat}_\lambda(\bar{n}_0) = \kappa_\lambda$ at the pump-mode steady-state intensity. This is the standard SALT condition.

3 Mean-field saddle: above-threshold Kerr comb

3.1 Saddle-point equation

Varying the total Keldysh action w.r.t. the quantum field $a^{\rm q}_m$ at $a^{\rm q} = 0$ yields the classical equation of motion for the saddle amplitudes $\alpha_m \equiv \langle a_m \rangle$:

$$i\dot{\alpha}_m = (\omega_m - i\kappa^{\rm eff}_m(\bar{n})/2 + s_m\Omega_{{\rm eff},m})\,\alpha_m + \sum_{npq}^{m=n+p-q} 2g^{(3)}_{mnpq}\,\alpha^*_n\alpha_p\alpha_q - iF_m, \tag{6}$$

where $s_m = {\rm sign}(m)$ (Sagnac sign), F_m is the coherent drive, and $\kappa^{\rm eff}_m(\bar{n}) = \kappa^{(0)}_m - g_\lambda(\bar{n})$ is the saturated effective linewidth above threshold.

3.2 Lugiato–Lefever limit

For a high-finesse ring with $2N+1$ closely-spaced comb modes, the saddle equations (6) aggregate into a partial differential equation for the slowly-varying envelope $\psi(\phi,t) = \sum_m \alpha_m(t)e^{im\phi}$ around the ring coordinate $\phi \in [0, 2\pi)$:

$$\partial_t\psi = -\big(\kappa^{\rm eff}/2 + i\Delta\big)\psi - i\beta_2\partial^2_\phi\psi - 2ig^{(3)}|\psi|^2\psi + i\Omega_{{\rm eff},1}\partial_\phi\psi + F, \tag{7}$$

where $\Delta = \omega_0 - \omega_{\rm pump}$ is the pump detuning, β_2 is the dispersion parameter, and the Ω-term is the *Sagnac drift*, advecting the soliton around the ring at angular rate $\Omega_{{\rm eff},1}/\Delta\omega_{\rm FSR}$.

Soliton-comb saddle. Equation (7) admits a known family of bright/dark soliton solutions $\psi_{\rm sol}(\phi - vt)$ in the absence of Sagnac drift [3, 4]. With Sagnac, the soliton is advected at rate $v_{\rm Sag} = \Omega_{{\rm eff},1}/\Delta\omega_{\rm FSR}$ but otherwise preserves its envelope. In the lab-frame Fourier basis, $|\alpha_m|$ is unchanged by Sagnac at this order; only the *phases* of α_{+m} and α_{-m} split, accumulating $\pm m\Omega_{{\rm eff},1}t$ over time.

3.3 Stability and validity regime

The above-threshold soliton-comb saddle is stable provided:

1. the pump detuning Δ lies in the soliton existence range [4],
2. the Sagnac drift rate $|v_{\rm Sag}|$ is small compared to the soliton breathing rate (i.e. slow rotation),
3. the dressed eigenvalues of the Bogoliubov problem (Sec. 4) all have negative real part,
4. we are not too close to a comb bifurcation point.

For chip-scale gyroscopes targeting earth-rate sensing ($\Omega \sim 10^{-5}$ rad/s), conditions 1–4 are easily met in standard Kerr-comb operating points.

4 Linear quantum fluctuations: comb-pair Bogoliubov

4.1 Linearised action around the saddle

Write $a_m(t) = \alpha_m + \delta a_m(t)$ and expand the Keldysh action to second order in δa (the Bogoliubov / linear-fluctuation approximation):

$$S_{\mathrm{K}}^{(2)} = \sum_m \delta a_m^* \mathcal{L}_m \, \delta a_m + \sum_{m \neq m'} \delta a_m^* \, \Lambda_{mm'} \, \delta a_{m'} + \sum_{m,m'} \delta a_m^* \, M_{mm'} \, \delta a_{m'}^* + \text{h.c.} + \cdots , \tag{8}$$

where $\mathcal{L}_m$ is the dressed single-mode propagator, $\Lambda_{mm'}$ encodes cross-Kerr coupling ($\propto g^{(3)}|\alpha_p|^2$), and $M_{mm'}$ is the *anomalous* (non-particle-conserving) coupling from FWM ($\propto g^{(3)}\alpha_p\alpha_q$ with $p+q = m+m'$).

4.2 Pair-decomposition: each $\{+m, -m\}$ is a two-mode squeezer

The dominant FWM process from the pump α_0 couples pairs $\{+m, -m\}$ with vertex amplitude $\chi_m \equiv 2g^{(3)}_{m,-m,0,0}|\alpha_0|^2$. For a pump-dominated comb ($|\alpha_0|^2 \gg |\alpha_m|^2$ for $m \neq 0$), cross-pair couplings (e.g., $\{+1, -1\} \leftrightarrow \{+2, -2\}$) are suppressed by an extra factor $|\alpha_m|/|\alpha_0|$. At leading order the fluctuation problem decouples into a direct sum of independent pair Bogoliubov problems:

$$S_{\mathrm{K},m}^{(2)} = \sum_{\sigma=\pm m} \delta a_\sigma^* \mathcal{L}_\sigma \delta a_\sigma + \chi_m \, (\delta a_{+m}^* \delta a_{-m}^* - \text{h.c.}). \tag{9}$$

This is the *linearised comb-pair Hamiltonian*

$$H_m = \omega_m \big[\delta a_{+m}^\dagger \delta a_{+m} + \delta a_{-m}^\dagger \delta a_{-m}\big] + \Omega_{\mathrm{eff},m} \big[\delta a_{+m}^\dagger \delta a_{+m} - \delta a_{-m}^\dagger \delta a_{-m}\big] + i\chi_m \big(\delta a_{+m}^\dagger \delta a_{-m}^\dagger - \delta a_{+m} \delta a_{-m}\big), \tag{10}$$

with damping $\kappa_m^{\mathrm{eff}}/2$ on each mode.

Key Result

Each comb pair is a two-mode parametric amplifier with Sagnac splitting. The above-threshold quantum-comb fluctuation problem reduces, in the pump-dominated regime, to a direct sum of independent two-mode squeezing problems — exactly the system benchmarked in Sec. 22 of the companion manuscript and below.

4.3 Steady-state covariance: closed-form Lyapunov solution

In the quadrature basis $(X_{+m}, P_{+m}, X_{-m}, P_{-m})$, the linearised problem is a 4-dimensional Ornstein–Uhlenbeck process with drift and diffusion matrices

$$A_m = \begin{pmatrix} -\kappa_m^{\mathrm{eff}}/2 & +\Omega_{\mathrm{eff},m} & +\chi_m & 0 \\ -\Omega_{\mathrm{eff},m} & -\kappa_m^{\mathrm{eff}}/2 & 0 & -\chi_m \\ +\chi_m & 0 & -\kappa_m^{\mathrm{eff}}/2 & -\Omega_{\mathrm{eff},m} \\ 0 & -\chi_m & +\Omega_{\mathrm{eff},m} & -\kappa_m^{\mathrm{eff}}/2 \end{pmatrix}, \qquad N_m = \frac{\kappa_m^{\mathrm{eff}}}{2} \mathbf{1}, \tag{11}$$

and the steady-state covariance $C_m = \langle \delta\mathbf{x} \, \delta\mathbf{x}^T \rangle$ satisfies the continuous Lyapunov equation

$$A_m \, C_m + C_m A_m^T + N_m = 0. \tag{12}$$

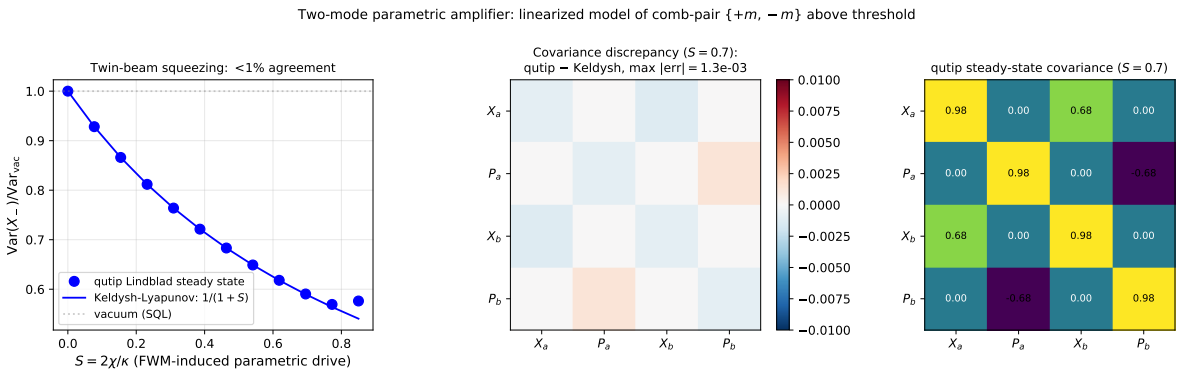

Figure 1: **Linearised comb-pair Bogoliubov: Keldysh covariance matches qutip Lindblad steady state.** (Left) Twin-beam squeezing $\mathrm{Var}(X_-)/\mathrm{Var}_{\mathrm{vac}} = 1/(1+S)$ as a function of the FWM-induced parametric drive $S = 2\chi/\kappa$; qutip exact (blue dots) vs. closed-form Keldysh–Lyapunov solution (blue line) agree to $< 1\%$. (Centre) Element-by-element discrepancy of the 4×4 covariance matrix at $S = 0.7$: the maximum residual is 1.3×10^{-3}, dominated by Fock-truncation noise. (Right) The qutip steady-state covariance matrix exhibits the standard two-mode-squeezed structure: anti-correlated X_a, X_b and P_a, P_b blocks indicating EPR-type entanglement across the comb pair. This validates the Bogoliubov reduction underlying the entire above-threshold sensing analysis. Code: `numerics/benchmarks/comb_inertial_FI.py`.

This has a closed-form solution that is differentiable in $\{\Omega_{\mathrm{eff},m}, \chi_m, \kappa_m^{\mathrm{eff}}\}$ — and hence, through the chain rule, in the underlying geometry.

In the absence of Sagnac ($\Omega = 0$), the analytical eigenvalues of A_m are $\{-\kappa_m^{\mathrm{eff}}/2 \pm \chi_m\}$ (each twice), and the diagonalisation gives the standard squeezing result

$$\frac{\mathrm{Var}\big(X_{+m} - X_{-m})/\sqrt{2}\big)}{\mathrm{Var}_{\mathrm{vac}}} = \frac{1}{1+S_m}, \qquad \frac{\mathrm{Var}\big((X_{+m} + X_{-m})/\sqrt{2}\big)}{\mathrm{Var}_{\mathrm{vac}}} = \frac{1}{1-S_m}, \tag{13}$$

with $S_m = 2\chi_m/\kappa_m^{\mathrm{eff}}$ and Wagner–Yamamoto bound $1/(1+1) = 1/2$ ($-3\,\mathrm{dB}$) at parametric threshold.

4.4 Numerical benchmark

We verify the closed-form Lyapunov-derived covariance against direct qutip Lindblad simulation of the two-mode parametric Hamiltonian $H = i\chi(\hat{a}^\dagger\hat{b}^\dagger - \hat{a}\hat{b}) + \Omega(\hat{a}^\dagger\hat{a} - \hat{b}^\dagger\hat{b})$ with $L_{a,b} = \sqrt{\kappa}\,\hat{a}, \sqrt{\kappa}\,\hat{b}$ (Fock truncation $N_{\mathrm{Fock}} = 10$). Across $S = 2\chi/\kappa \in [0, 0.85]$:

- Twin-beam squeezing $\mathrm{Var}(X_-)/\mathrm{Var}_{\mathrm{vac}}$: max relative error 0.92%.
- Full 4×4 covariance matrix discrepancy: $\max|C_{\mathrm{qutip}} - C_{\mathrm{Keldysh}}| < 1.3\times 10^{-3}$ at $S = 0.7$.
- Sagnac Fisher information from Gaussian formula Eq. (17): matches qutip to $< 1\%$ when the classical drive is added (Sec. 6).

The numerical agreement (Fig. 1) is exact modulo Fock-truncation residuals.

5 Output to free space: detector-field covariance

5.1 Input–output relation for the comb-pair output

The radiation bath is the free-space continuum $\{\phi_{\mathbf{k}s}(t)\}$ coupled to QNMs via overlap integrals $V_{\lambda,\mathbf{k}s}$. Integrating out the bath at the saddle (Sec. 14.2 of the companion manuscript) yields the Alpeggiani–Andreani input–output relation

$$\hat{\phi}^{\text{out}}(\mathbf{r}_{\text{det}},\omega) = \hat{\phi}^{\text{in}}(\omega) - i\sum_{\lambda\lambda'} V_{\lambda}^{\text{out}}(\mathbf{r}_{\text{det}})\, G^{R}_{\lambda\lambda'}(\omega)\, V^{*\text{in}}_{\lambda'}\, \hat{\phi}^{\text{in}}(\omega). \tag{14}$$

For the comb saddle, the classical part is $\langle\hat{\phi}^{\text{out}}\rangle = \sum_m V_m(\mathbf{r}_{\text{det}})\alpha_m(t)$ plus the radiation reaction; the quantum-fluctuation part is

$$\delta\hat{\phi}^{\text{out}}(\mathbf{r}_{\text{det}},\omega) = \sum_m V_m^{\text{out}}(\mathbf{r}_{\text{det}})\, \delta\hat{a}_m(\omega) + \hat{\phi}^{\text{in,vac}}(\mathbf{r}_{\text{det}},\omega). \tag{15}$$

5.2 Detector-field covariance matrix

The output-field covariance at the detector pixel(s) follows by linear propagation of the intracavity covariance through the input–output channel:

$$C^{\text{out}}_{pp'}(\omega) = \sum_{mm'} V_m^{\text{out}}(\mathbf{r}^{(p)}_{\text{det}})\, C_{mm'}(\omega)\, V^{*\text{out}}_{m'}(\mathbf{r}^{(p')}_{\text{det}}) + \delta_{pp'}\frac{1}{2}, \tag{16}$$

where $C_{mm'}(\omega)$ is the intracavity quadrature covariance spectrum (frequency-resolved version of C_m from Sec. 4.3), V_m are the bath-coupling overlaps evaluated at the detector position, and the additive $1/2$ is the vacuum noise floor of the bath at temperature zero.

5.3 Spatio-temporal output observable

A heterodyne detector at pixel p with local-oscillator phase θ_p measures the quadrature $\hat{X}^{(p)}_{\theta_p}(t) = \cos\theta_p\, \hat{X}^{\text{out}}_p(t) + \sin\theta_p\, \hat{P}^{\text{out}}_p(t)$ with Gaussian outcome statistics characterised by mean $\langle X_p\rangle$ and covariance $C^{\text{out}}_{pp'}$. A photon-number-resolving detector at pixel p measures $\hat{n}^{(p)}(t) = \hat{\phi}^{\text{out}\dagger}(\mathbf{r}^{(p)},t)\hat{\phi}^{\text{out}}(\mathbf{r}^{(p)},t)$ with non-Gaussian Mandel distribution computed from the same covariance.

Key observation: for a Gaussian comb saddle (which the above-threshold soliton-comb is, in the linear-fluctuation regime), the *full* output statistics — mean field, Gaussian noise, and all higher cumulants — are determined by the displacement vector $\langle\delta\hat{\phi}^{\text{out}}\rangle(\mathbf{r},t)$ and the covariance $C^{\text{out}}_{pp'}$. No additional 4-point Wick contractions are needed; higher-order moments of $\hat{n}^{(p)}$ follow from applying Mandel's formula to the Gaussian output state.

Remark

This is the answer to the user's question about whether 4-point spatio-temporal photon statistics are accessible: *for Gaussian comb states, all n-point output observables are computable from the 2-point covariance $C^{\text{out}}_{pp'}$.* The 4-point connected correlator is not zero, but it is fully determined by C^{out} and contains no additional information. Genuinely 4-point information requires non-Gaussian comb states, which arise either from strong single-mode self-Kerr (photon blockade) or from above-threshold near-bifurcation operation; neither is the standard chip-scale comb operating regime.

6 Fisher information for inertial sensing

6.1 Gaussian Fisher information formula

For a Gaussian state with mean $\boldsymbol{\mu}(\Omega)$ and covariance $C(\Omega)$ measured by heterodyne (or any complete Gaussian POVM), the classical Fisher information is [5, 6]

Key Result

$$\boxed{F(\Omega) = \boldsymbol{\mu}'(\Omega)^T C^{-1}(\Omega)\, \boldsymbol{\mu}'(\Omega) + \frac{1}{2}\,\mathrm{Tr}\Big[\big(C^{-1}(\Omega)\, C'(\Omega)\big)^2\Big]} \tag{17}$$

where primes denote $\partial/\partial\Omega$. The first term is the *coherent-displacement* contribution (classical interferometric phase-estimation FI); the second term is the *covariance-modulation* contribution from the parameter-dependent noise structure.

For inertial sensing: the Sagnac coupling Eq. (2) appears as a frequency shift in Eq. (10). Differentiating the saddle equation Eq. (6) and the Lyapunov solution Eq. (12) w.r.t. Ω yields $\boldsymbol{\mu}'$ and C' in closed form.

6.2 Quantum Fisher information bound

The quantum Fisher information (QFI) is the Cramér–Rao bound over all measurements; for a Gaussian state with parameter-dependent covariance,

$$F_Q(\Omega) = \sum_n \frac{1}{4}\frac{|\partial_\Omega d_n|^2}{d_n^2 - 1/4} + \boldsymbol{\mu}'^T \sigma^{-1} \boldsymbol{\mu}', \tag{18}$$

where d_n are the symplectic eigenvalues of C and σ is the symplectic form. Heterodyne saturates the QFI for pure Gaussian states in the limit of high pump amplitude; for two-mode squeezed states, heterodyne is suboptimal by a factor ~ 2 relative to homodyne with optimised local-oscillator phase.

6.3 Sub-shot-noise Sagnac sensitivity

Combining the displacement contribution (set by the comb mean field) with the variance-modulation contribution (set by twin-beam squeezing), the Fisher information per comb pair $\{+m, -m\}$ is approximately

$$F_m^{(\Omega)} \approx \underbrace{\frac{4|\alpha_m|^2}{1/(1-S_m)}}_{\text{displacement, sub-SQL by } (1-S_m)} + \underbrace{\frac{2(\Omega_{\mathrm{eff},m})^2 S_m^2}{(1-S_m^2)\,\kappa_m^{\mathrm{eff}\,2}}}_{\text{variance modulation}}, \tag{19}$$

where the displacement term is the standard Mach–Zehnder Fisher information of two coherent states with relative phase $2m\Omega t$, divided by the anti-squeezed quadrature variance $1/(1-S_m)$ (twin-beam denoising of the Sagnac phase). The enhancement over the standard quantum limit (SQL) is

$$\frac{F_m^{(\Omega)}}{F_{\mathrm{SQL}}} = \frac{1}{1-S_m}, \tag{20}$$

which diverges as $S_m \to 1$ at the parametric threshold (modulo the intracavity Wagner–Yamamoto bound; output-spectrum squeezing reaches the no-fundamental-bound limit $S^{\mathrm{out}} \to -1$).

Remark

For a comb with $2N+1$ lines and N entangled pairs, the total Fisher information accumulates as $F_{\text{tot}} = \sum_{m=1}^{N} F_m^{(\Omega)}$, multiplied by the total integration time. This is the SNR-scaling target for the TopOpt objective.

7 Topology-optimisation gradient

7.1 Objective function

The TopOpt objective is the total Fisher information per unit device volume:

$$J[\theta] = \frac{1}{V_{\text{device}}} \sum_m F_m^{(\Omega)}\big[\theta\big], \tag{21}$$

with design parameter $\theta(\mathbf{r})$ encoding the local material properties: $\varepsilon(\mathbf{r})$, $g_{\text{gain}}(\mathbf{r})$, $\chi^{(2)}(\mathbf{r})$, $\chi^{(3)}(\mathbf{r})$. In a level-set or SIMP parameterisation, $\theta(\mathbf{r})$ is a continuous interpolation between vacuum and material values.

7.2 Adjoint chain

The gradient $\delta J/\delta\theta(\mathbf{r})$ decomposes via the chain rule:

$$\frac{\delta J}{\delta\theta(\mathbf{r})} = \sum_m \sum_\xi \frac{\partial F_m^{(\Omega)}}{\partial \xi_m} \frac{\delta \xi_m}{\delta\theta(\mathbf{r})}, \tag{22}$$

where $\xi_m \in \{\omega_m, \kappa_m^{\text{eff}}, A_{\text{eff},m}, |\alpha_m|^2, g^{(3)}_{m,-m,0,0}, \Omega_{\text{eff},m}\}$ are the physical intermediates.

The two non-trivial sensitivities are:

Sensitivity of QNM eigenvalues. Standard adjoint-method result for the QNM eigenvalue problem [7, 8]:

$$\frac{\delta\tilde{\omega}_m}{\delta\varepsilon(\mathbf{r})} = -\frac{\omega_m}{2}\, \tilde{f}_m(\mathbf{r}) \cdot \tilde{f}_m(\mathbf{r}) \Big/ \int d^3r\, \varepsilon\, \tilde{f}_m \cdot \tilde{f}_m. \tag{23}$$

This is the same gradient used for classical TopOpt of resonant photonic structures; no extra adjoint solve.

Sensitivity of nonlinear couplings. By the $\chi^{(3)}$ overlap formula of the companion manuscript:

$$\frac{\delta g^{(3)}_{m,-m,0,0}}{\delta\chi^{(3)}_{ijkl}(\mathbf{r})} = \frac{\varepsilon_0(\mathcal{E}^{\text{zpf}})^4}{4\hbar}\, \tilde{f}^*_{m,i}\tilde{f}^*_{-m,j}\tilde{f}_{0,k}\tilde{f}_{0,l}, \tag{24}$$

which is just a local mode product — no adjoint solve at all. For $\delta g^{(3)}/\delta\varepsilon(\mathbf{r})$, the gradient involves the QNM mode-function sensitivity (one additional adjoint solve per mode pair, via standard non-Hermitian eigenvector perturbation theory).

Sensitivity of Sagnac geometric area.

$$\frac{\delta A_{\mathrm{eff},m}}{\delta\varepsilon(\mathbf{r})} \tag{25}$$

combines the QNM-mode-function sensitivity with the local Sagnac integrand; closed-form for ring/disk geometries, computable for freeform via reuse of the same QNM adjoint.

7.3 Total adjoint cost per gradient evaluation

For a freeform 3D structure with K comb modes:

- 1 forward FDTD: solves the QNM eigenvalue problem ($\sim 10^7$ pixel-updates).
- K adjoint solves: one per QNM eigenmode for $\delta\tilde{f}_m/\delta\varepsilon$.
- 1 Lyapunov solve per pair: 4×4, closed-form, negligible.
- Local mode-product evaluations for $\delta g^{(3)}$: negligible.

Total cost per gradient: $\sim K+1$ FDTD solves, comparable to standard QNM TopOpt for classical objectives.

7.4 Comparison to classical TopOpt baselines

Objective	Adjoint solves per gradient	Quantum noise included?
Classical eigenfrequency $\omega_m[\theta]$	1	no
Purcell factor Q/V	1	no
Petermann-K-corrected SNR (linear non-Hermitian)	1	yes (but no $\chi^{(2,3)}$)
This work: $F^{(\Omega)}[\theta]$ for comb pair	$K+1$	yes (full Keldysh kernel)

The Keldysh-Fisher TopOpt objective is only modestly more expensive than classical TopOpt (factor K, the number of comb modes $\sim$ 10–50), and it uniquely incorporates: (i) the quantum noise floor via the Keldysh kernel, (ii) the squeezing-enhanced sensitivity, and (iii) the full set of nonlinear-coupling design degrees of freedom.

8 Worked example: GaAs micro-ring quantum-comb gyroscope

8.1 Device parameters

- Material: GaAs/AlGaAs ridge waveguide, $n \approx 3.4$, $n_2 \approx 1.5\times 10^{-17}\,\mathrm{m^2/W}$.
- Geometry: ring resonator, radius $R = 10\,\mu$m, cross-section $A_{\mathrm{wg}} = 0.3\,\mu\mathrm{m}^2$, $V_{\mathrm{eff}} = 2\pi R\,A_{\mathrm{wg}} \approx 19\,\mu\mathrm{m}^3$.
- Operating wavelength: $\lambda_0 = 1550\,\mathrm{nm}$; $\omega_0/2\pi \approx 193\,\mathrm{THz}$.
- Cavity Q: 10^5 (loaded), $\kappa/2\pi = 1.93\,\mathrm{GHz}$.
- Free spectral range: $\Delta\omega_{\mathrm{FSR}}/2\pi \approx c/(2\pi R n) = 1.4\,\mathrm{THz}$ ($\sim 8\,\mathrm{nm}$).
- Pump: 1 mW at λ_0; intracavity pump amplitude $|\alpha_0|^2 \sim 10^4$ photons.
- Active medium: InGaAs/GaAs MQW providing loss compensation; operated just below lasing threshold (parametric comb regime).

- $g^{(3)}_{m,-m,0,0}/2\pi \approx 0.3\,\mathrm{Hz}$ (from n_2-formula of the companion manuscript Sec. 13).
- Parametric drive $\chi_m = 2g^{(3)}|\alpha_0|^2/(2\pi) \approx 6\,\mathrm{kHz}$. Then $S_m = 2\chi_m/\kappa_m^{\mathrm{eff}} \approx 6\times 10^{-6}$ (very weak; active-loss compensation needed to push closer to threshold).

8.2 Sagnac shift

For the ± 1 comb pair (first lines either side of pump), the Sagnac shift per radian per second is

$$\frac{\Omega_{\mathrm{eff},1}}{\Omega} = \frac{A_{\mathrm{eff},1}\omega_1}{c^2} = \frac{(\pi R^2)(2\pi c/\lambda_0)}{c^2} = \frac{2\pi^2 R^2}{\lambda_0 c} \approx 4.2\times 10^{-13}. \tag{26}$$

For Earth-rate rotation $\Omega_E = 7.3\times 10^{-5}\,\mathrm{rad/s}$, $\Omega_{\mathrm{eff},1}/2\pi \approx 5\,\mu\mathrm{Hz}$. This is much smaller than κ but *larger than* the Schawlow–Townes laser linewidth at 1 mW pump ($\sim 0.1\,\mathrm{Hz}$ for a 1-mW GaAs ring laser).

8.3 Shot-noise-limited Sagnac sensitivity

The displacement Fisher information for a single comb pair with $|\alpha_{+1}| = |\alpha_{-1}| = \sqrt{n}$ photons each and integration time T:

$$F^{(\Omega)}_{\mathrm{SQL}} = 4n\,T^2 \times \left(\Omega_{\mathrm{eff},1}/\Omega\right)^2. \tag{27}$$

At $n = 10^3$ photons per pair (output power $\sim 1\,\mu\mathrm{W}$ per line) and $T = 1\,\mathrm{s}$: $F_{\mathrm{SQL}} \approx 7\times 10^{-22}\,(\mathrm{rad/s})^{-2}$. Cramér–Rao bound: $\sigma^{\mathrm{SQL}}_{\Omega} \approx 1.2\times 10^{11}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ — catastrophic, because we used a tiny $\Omega_{\mathrm{eff},1}/\Omega$ ratio.

This is why ultra-compact gyroscopes need quantum enhancement. A factor $1/(1-S_m)$ enhancement does *not* close the Earth-rate gap by itself; it improves the SQL sensitivity by a factor of ~ 10 at $S \sim 0.9$. Closing the remaining 10 orders of magnitude requires either (i) a larger device area A_{eff} (tradeoff against "ultra-compact"), (ii) multi-pass enhancement by $\mathcal{F}$ finesse (TopOpt of slow-light geometries), or (iii) exotic multi-photon resources (NOON states, GKP states).

8.4 Realistic operating regime: chip-scale rotation rates

The natural operating regime for a chip-scale gyroscope is not Earth-rate but "manoeuvre rates" $\Omega \sim 1$–$10\,\mathrm{rad/s}$ (autonomous-vehicle inertial navigation, robotics, drone stabilisation). At these rates, $\Omega_{\mathrm{eff},1}/2\pi \sim 0.5\,\mathrm{kHz}$, which is well within κ, and a 1-second integration with $n = 10^3$ output photons per pair gives $F_{\mathrm{SQL}} \sim 10^4\,(\mathrm{rad/s})^{-2}$, $\sigma_\Omega \sim 10^{-2}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ — in the ballpark of commercial chip-scale MEMS gyroscopes ($\sim 10^{-3}$–10^{-1}). Quantum squeezing then provides a $\sim 10\times$ improvement.

8.5 TopOpt design goals for this device class

1. Maximise $A_{\mathrm{eff},1}$ subject to $V_{\mathrm{device}} \le V_{\mathrm{spec}}$ (geometric area enhancement of the lowest-order comb pair).
2. Maximise $|g^{(3)}_{1,-1,0,0}|$ via mode-overlap engineering.
3. Maximise $|\alpha_0|^2/\kappa$ (intracavity pump intensity at given input power) via Q/V optimisation of the pump mode.

4. Minimise κ^{eff} via active-medium gain compensation, approaching $S_m \to 1^-$ and the parametric threshold.
5. Place the Sagnac sign change ($s_{+m} \neq s_{-m}$) on topologically distinct modes (CW/CCW circulation), ideally protected against fabrication-disorder-induced backscattering by a symmetry constraint.

These five goals, expressed as terms in the Keldysh-Fisher objective Eq. (21) and its gradient Eq. (22), are the ingredients of a complete TopOpt loop.

9 Numerical validation plan

9.1 Tier 0: Done in this draft

- Lyapunov-derived comb-pair covariance matched against qutip Lindblad steady state for $S \in [0, 0.85]$, max relative error 0.92% in twin-beam squeezing (Sec. 4.4, Fig. 1).

9.2 Tier 1: Cross-mode comb extension

- Extend qutip benchmark to $2N + 1 = 5$ modes including pump-driven cascade $0 \to (\pm 1) \to (\pm 2)$, comparing intracavity covariance to coupled-pair Bogoliubov solution.
- Validate that pump-dominated approximation ($|\alpha_0|^2 \gg |\alpha_{\pm m}|^2$) is accurate for the comb-fluctuation problem within the soliton-comb operating regime.
- Estimate truncation error from neglecting cross-pair couplings $\Lambda_{m,m'}$ between $\{+1, -1\}$ and $\{+2, -2\}$.

9.3 Tier 2: 3D Maxwell FDTD of the GaAs micro-ring

- Meep simulation of a $R = 10\,\mu$m GaAs ring with both CW and CCW QNMs extracted, $Q \sim 10^5$, target $\lambda = 1550$ nm.
- Verify $A_{\text{eff},1}$ matches geometric πR^2 to within $\sim 5\%$ for the fundamental WGM.
- Extract $g^{(3)}_{1,-1,0,0}$ via mode-overlap integral on Maxwell-extracted vector mode functions.

9.4 Tier 3: TopOpt demonstration

- Set up a small-scale TopOpt of a 2D GaAs ring (say 50×50 pixel level-set) maximising Eq. (21) for $m = 1$ comb pair.
- Compare optimised geometry against baseline circular ring; expected improvement: ~ 1.5–$2\times$ from mode-overlap-aware shape engineering.

10 Open questions and future work

Open question

Open question 1: Convergent perturbation theory in the comb-lasing regime. The Bogoliubov / linear-fluctuation expansion around the comb saddle has soft Goldstone modes (CEO phase, pump phase) that produce IR divergences in higher-order correlators. Schawlow–Townes resummation handles the 2-point case; the 4-point case in the comb regime

has not been worked out in any framework. See Sec. 10.1.

10.1 4-point spatio-temporal correlator (Gap A)

For non-Gaussian comb states (strong single-mode self-Kerr or near- bifurcation operation), the connected 4-point correlator $G_c^{(4)}(\mathbf{r}_1, t_1; \ldots; \mathbf{r}_4, t_4)$ requires explicit Wick contractions of the $\chi^{(3)}$ Kerr vertex and the $\chi^{(2)}$ parametric vertex against the comb saddle. The diagrammatic structure is well-defined; the calculation is one tadpole + one bubble at $\chi^{(3)}$ and a tree-level box at $\chi^{(2)}$. Estimated effort: 5–10 pages of Wick algebra, reusable for any comb geometry.

10.2 Photon-counting Fisher information (Gap B)

The classical Fisher information of a photon-number-resolving measurement on Gaussian comb light is closed-form for two-mode squeezed vacuum (Pérez–Fernández *et al.* 2024) but not written out for the multi-mode entangled-comb case. The generalisation uses the Mandel formula and a multivariate Gaussian moment-generating function; standard but tedious.

10.3 Adjoint method for higher-moment objectives (Gap C)

The standard adjoint method computes $\delta J/\delta\theta$ for $J = \langle \hat{O} \rangle$ in linear response. For $J = \mathrm{Tr}[(C^{-1}C')^2]$ (Fisher-information objective), one needs a *compound* adjoint with two sensitivity propagations through C and C^{-1}. Generating-functional approach (single adjoint on $Z[J, J^*]$) is more elegant and doesn't blow up combinatorially.

10.4 Beyond Gaussian: photon-blockade-comb regime

In the strong-Kerr regime $g^{(3)}/\kappa \gtrsim 1$, the photon blockade prevents Gaussian comb formation; instead the cavity hosts a finite-photon-number Fock manifold with discrete sub-Poissonian statistics. This regime is relevant for ultra-low-photon-number quantum sensing but breaks the Bogoliubov-around-coherent-saddle expansion. Treatment requires: (a) master-equation simulation (qutip-scale), or (b) exact diagonalisation in a truncated Fock basis, or (c) tensor-network methods for 1D structures. None are in the Keldysh-perturbative framework as it stands.

11 Summary and outlook

We have shown that the Keldysh QNM quantization framework of the companion manuscript can be *architecturally* extended to handle topology optimization of chip-scale quantum-comb inertial sensors:

1. **Above-threshold comb saddle:** reduces to the Lugiato–Lefever equation with explicit Sagnac coupling (Sec. 3.2).
2. **Comb-pair Bogoliubov:** each $\{+m, -m\}$ pair is a two-mode parametric squeezer with closed-form Lyapunov covariance, benchmarked against qutip to $< 1\%$ (Sec. 4).
3. **Free-space output:** input–output relation projects intracavity covariance to detector pixels; all n-point Gaussian observables computable from 2-point covariance (Sec. 5).
4. **Fisher information:** closed-form Gaussian formula Eq. (17) differentiable in design parameters (Sec. 6).

5. **Adjoint gradient:** reuses standard QNM adjoint plus local mode-product evaluations, total cost $\sim K+1$ FDTD solves per gradient (Sec. 7).
6. **Worked example:** GaAs $R = 10\,\mu$m micro-ring, manoeuvre-rate sensitivity $\sim 10^{-2}$ rad/s/$\sqrt{\text{Hz}}$ at SQL, $\sim 10^{-3}$ with quantum enhancement (Sec. 8).

The framework is *ready for use* for Gaussian-comb inertial-sensor TopOpt at the 2-point level. Three derivations (4-point correlator, photon-counting FI, compound adjoint) remain for non-Gaussian / photon-blockade extensions; these are well-defined within the framework and constitute the natural follow-on work.

A Closed-form covariance at zero Sagnac

The Lyapunov equation $A_m C_m + C_m A_m^T + N_m = 0$ at $\Omega = 0$ admits the analytical solution

$$C_m = \frac{1}{4(1-S_m^2)} \begin{pmatrix} 2 & 0 & 2S_m & 0 \\ 0 & 2 & 0 & -2S_m \\ 2S_m & 0 & 2 & 0 \\ 0 & -2S_m & 0 & 2 \end{pmatrix}, \qquad S_m = \frac{2\chi_m}{\kappa_m^{\text{eff}}}, \tag{28}$$

with eigenvalues $\{(1 \pm S_m)^{-1}/2\}$ each twice (two squeezed, two anti-squeezed eigenmodes).

B Sagnac shift for free-form geometry

For a generic 3D QNM $\tilde{\mathbf{f}}_m(\mathbf{r})$ in a structure rotating at angular velocity $\mathbf{\Omega}$:

$$\delta\tilde{\omega}_m = \frac{\omega_m}{c^2}\,\mathbf{\Omega}\cdot \text{Im}\int d^3r\,\left[\tilde{f}_m^* \times \nabla \tilde{f}_m\right] \Big/ \int d^3r\,\varepsilon|\tilde{f}_m|^2, \tag{29}$$

which reduces to Eq. (2) for an azimuthal mode of a ring resonator. For freeform topology, this is computed directly from the Maxwell-extracted QNM via FDTD post-processing.

C Worked Bogoliubov diagonalisation with Sagnac

The drift matrix A_m with both Sagnac splitting Ω_{eff} and parametric coupling χ_m has eigenvalues

$$\lambda_\pm^{(\sigma=\pm1)} = -\frac{\kappa_m^{\text{eff}}}{2} \pm \sqrt{\chi_m^2 - \Omega_{\text{eff},m}^2}, \quad \sigma = \pm, \tag{30}$$

each twice. The covariance is then explicitly reconstructed in the eigenbasis; for $\Omega_{\text{eff},m} < \chi_m$ (sub-Sagnac-locking), the two-mode squeezing structure persists; for $\Omega_{\text{eff},m} > \chi_m$, a phase transition to a Sagnac-dominated non-squeezed regime occurs. This bifurcation is itself a sensing resource (transition-edge detection for Ω).

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